

SOLUTION:

- Consider a system consisting of the two gear. Noting that the gear rotational speeds are related, evaluate the final kinetic energy of the system.

$$\omega_B = \frac{(600 \text{ rpm})(2\pi \text{ rad/rev})}{60 \text{ s/min}} = 62.8 \text{ rad/s}$$

$$\omega_A = \omega_B \frac{r_B}{r_A} = 62.8 \frac{0.100}{0.250} = 25.1 \text{ rad/s}$$

$$I_A = m_A k_A^2 = (10 \text{ kg})(0.200 \text{ m})^2 = 0.400 \text{ kg} \cdot \text{m}^2$$

$$I_B = m_B k_B^2 = (3 \text{ kg})(0.080 \text{ m})^2 = 0.0192 \text{ kg} \cdot \text{m}^2$$

$$T_2 = \frac{1}{2} I_A \omega_A^2 + \frac{1}{2} I_B \omega_B^2 = \frac{1}{2} (0.400)(25.1)^2 + \frac{1}{2} (0.0192)(62.8)^2 = 163.9 \text{ J}$$

SOLUTION:

- The weight and spring forces are conservative. The principle of work and energy can be expressed as $T_1 + V_1 = T_2 + V_2$.

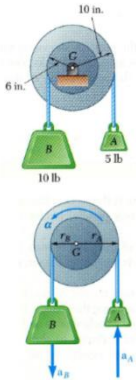
- Evaluate the initial and final potential energy.

$$V_1 = V_g + V_e = 0 + \frac{1}{2} kx_1^2 = \frac{1}{2} (1800 \text{ lb/in.})(1 \text{ in.})^2 = 900 \text{ in} \cdot \text{lb} = 75 \text{ ft} \cdot \text{lb}$$

$$V_2 = V_g + V_e = Wh + 0 = (30 \text{ lb})(1.5 \text{ ft}) = 45 \text{ ft} \cdot \text{lb}$$

- Express the final kinetic energy in terms of the angular velocity of the rod.

$$T_2 = \frac{1}{2} m v_G^2 + \frac{1}{2} I_G \omega_2^2 = \frac{1}{2} m (r \omega_2)^2 + \frac{1}{2} I_G \omega_2^2 = \frac{1}{2} (30) (1.5 \omega_2)^2 + \frac{1}{2} (1.941) \omega_2^2 = 2.019 \omega_2^2$$



SOLUTION:

- Determine the direction of rotation by evaluating moment on the pulley due to the two blocks.

$$+\circlearrowleft \sum M_G = (10 \text{ lb})(6 \text{ in}) - (5 \text{ lb})(10 \text{ in}) = 10 \text{ in} \cdot \text{lb}$$

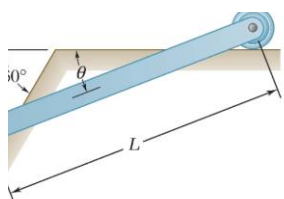
rotation is counterclockwise.

$$\text{note: } I = m k^2 = \frac{W}{g} k^2 = \frac{12 \text{ lb}}{32.2 \text{ ft/s}^2} \left(\frac{8}{12} \text{ ft} \right)^2 = 0.1656 \text{ lb} \cdot \text{ft} \cdot \text{s}^2$$

- Relate the acceleration of the block and the acceleration of the pulley.

$$a_A = r_A \alpha = \left(\frac{10}{12} \text{ ft} \right) \alpha \quad a_B = r_B \alpha = \left(\frac{6}{12} \text{ ft} \right) \alpha$$

$$T = \frac{1}{2} m v_G^2 + \frac{1}{2} I_G \omega^2 \quad s_B = L \sin \theta$$



$$\Delta y_B = s_B \sin 60^\circ = L \sin \theta \cdot \frac{\sqrt{3}}{2}$$

$$\Delta y_G = \frac{1}{2} \Delta y_B = \frac{L \sqrt{3}}{4} \sin \theta$$

几何形状

转动惯量公式

矩形 (Rectangle)

$$I'_x = \frac{1}{12} b h^3$$

$$I'_y = \frac{1}{12} b h^3$$

$$I_y = \frac{1}{3} b h^3$$

$$J_C = \frac{1}{12} b h (b^2 + h^2)$$

三角形 (Triangle)

$$I'_x = \frac{1}{36} b h^3$$

$$I_x = \frac{1}{12} b h^3$$

半圆形 (Semicircle)

$$I_x = I_y = \frac{1}{8} \pi r^4$$

$$J_O = \frac{1}{4} \pi r^4$$

四分之一圆形 (Quarter Circle)

$$I_x = I_y = \frac{1}{16} \pi r^4$$

$$J_O = \frac{1}{8} \pi r^4$$

圆形 (Circle)

$$I_x = I_y = \frac{1}{4} \pi r^4$$

$$J_O = \frac{1}{2} \pi r^4$$

椭圆形 (Ellipse)

$$I_x = \frac{1}{4} \pi a b^3$$

$$I_y = \frac{1}{4} \pi a b^3$$

$$J_O = \frac{1}{4} \pi a b (a^2 + b^2)$$

Define moment of inertia

$$I_{G,x} = \int (y^2 + z^2) dm$$

$$I_{G,xy} = \int xy dm$$

$$I_{G,xz} = \int xz dm$$

$$I_{G,yz} = \int yz dm$$

$$\mathbf{L} = \int \mathbf{r} \times \mathbf{v} dm$$

$$\mathbf{L} = I \boldsymbol{\omega}$$

17.146 A large 3-lb sphere with a radius $r = 3$ in. is thrown into a light basket at the end of a thin, uniform rod weighing 2 lb and length $L = 10$ in. as shown. Immediately before the impact the angular velocity of the rod is 3 rad/s counterclockwise and the velocity of the sphere is 2 ft/s down. Assume the sphere sticks in the basket. Determine after the impact (a) the angular velocity of the basket and sphere, (b) the components of the reactions at A.

Conservation of Angular Momentum

$$\frac{1}{2} m_r L^2 \omega_0 - L m_s v_0 = \left(\frac{1}{3} m_r L^2 + \frac{2}{5} m_s r^2 + m_s (r^2 + L^2) \right) \omega$$

$$\omega = -1.286 \text{ rad/s} \quad \text{后: 纯转动, 小球}$$

$$\text{前: 平动, 质心列辅助量}$$

$$\sum M = I_G \alpha$$

$$m_r g \frac{L}{2} + m_s g L = \left(\frac{1}{3} m_r L^2 + \frac{2}{5} m_s r^2 + m_s (r^2 + L^2) \right) \alpha$$

$$\alpha = 38.21 \text{ rad/s}^2$$

$$a_{rr} = \alpha \frac{L}{2} = \frac{1}{2} (38.21) (10) = 191.05 \text{ ft/s}^2$$

$$a_{rt} = \alpha r = (38.21) (3) = 114.63 \text{ ft/s}^2$$

$$a_{rt} = \omega^2 \frac{L}{2} = (1.286)^2 \frac{10}{2} = 8.34 \text{ ft/s}^2$$

$$a_{rt} = \omega^2 r = (1.286)^2 (3) = 4.98 \text{ ft/s}^2$$

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16.78 A uniform slender rod of length $L = 900$ mm and mass $m = 4$ kg is suspended from a hinge at C. A horizontal force P of magnitude 75 N is applied at end B. Knowing that $\bar{r} = 225$ mm, determine (a) the angular acceleration of the rod, (b) the components of the reaction at C.

$$M = I \alpha$$

$$P \left(\frac{L}{2} + \bar{r} \right) = \left(\frac{1}{12} m L^2 + m \bar{r}^2 \right) \alpha$$

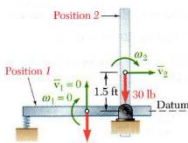
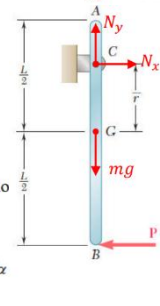
$$\alpha = \frac{P \left(\frac{L}{2} + \bar{r} \right)}{m \left(\frac{1}{12} L^2 + \bar{r}^2 \right)} = 107.1 \text{ rad/s}^2$$

$$N_x - P = m a_{Gx} = -m \bar{r} \alpha$$

$$N_x = -21.4 \text{ N} \leftarrow$$

$$N_y - m g = m a_{Gy} = 0$$

$$N_y = 39.2 \text{ N} \uparrow$$



$$I = \frac{1}{12} m L^2$$

$$= \frac{1}{12} \left(\frac{30 \text{ lb}}{32.2 \text{ ft/s}^2} \right) (5 \text{ ft})^2$$

$$= 1.941 \text{ lb} \cdot \text{ft} \cdot \text{s}^2$$

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Two hemispheres are held together by a spring under compression (the spring is 1 spheres). The potential energy of the compressed spring is 120 J and the assembly has an initial velocity v_0 of magnitude $v_0 = 8$ m/s. Knowing that the cord is severed when $\theta = 30^\circ$, causing the hemispheres to fly apart, determine the resulting velocity of each hemisphere.

Momentum conservation

$$m_A v_A = m_B v_B \quad (1)$$

$$\vec{v}_A = \vec{v}_0 + \vec{v}'_A$$

$$\vec{v}_B = \vec{v}_0 + \vec{v}'_B$$

Energy conservation

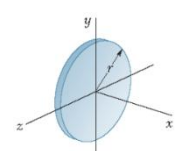
$$E_p + \frac{1}{2} (m_A + m_B) v_0^2 = \frac{1}{2}$$

$$v_A^2 = v_0^2 + v_A'^2 - 2 v_0 v_A' \cos 30^\circ \quad (3)$$

$$v_B^2 = v_0^2 + v_B'^2 - 2 v_0 v_B' \cos 30^\circ \quad (4)$$

$$v_A' = 6 \text{ m/s} \quad v_A = 4.11 \text{ m/s} \quad \angle 46.88^\circ$$

$$v_B' = 10 \text{ m/s} \quad v_B = 17.39 \text{ m/s} \quad \angle 16.71^\circ$$



$$I_x = \frac{1}{2} m r^2$$

$$I_y = I_z = \frac{1}{4} m r^2$$

